

Jacobian-lens trajectories under medical SFT

`gemma-3-{270m,1b,4b,12b,27b}-it`

August 10, 2026

Full fine-tune on `mix_default`, `lr 1e-5`, log-spaced checkpoints to step 4882

Five arms complete; findings 2–3 are settled. All behavioral numbers are on the **current** instrument (`rp=1.1`, self-hosted Kimi judge); nothing here is comparable to a version of this deck dated before 2026-08-05.

Setup

- **Five arms**, identical seed and data order: gemma-3-{270m,1b,4b,12b,27b}-it, full FT, lr $1e-5$, completion-only loss.
- **Log-spaced checkpoints** $\{1, 2, 4, \dots, 4096\}$ + final (4882). $t = 0$ is the untouched base, prepended by the probe.
- **Lens refit per checkpoint** on a frozen 1000-sequence generic (non-medical) WikiText corpus, taken from jlens's own load_wikitext_prompts — the lens is a function of the weights.
- **Concordance** on the frozen 193-prompt probe set: 50 general + 50 medical (hand-authored one-hop cloze, style-matched control pairs) + 93 multihop from jlens's lens-eval-multihop.json.
- **Behavioral** trajectory in parallel: MedQA, MedMCQA, PubMedQA, reported as *answered* accuracy with answer rate alongside. **Scored by an LLM judge**, not a regex — see the instrument slide.

All lens numbers below are the **mean over the last $n/5$ source layers** (the probe's `kl_final` / `top1_final`): 3 layers at 270m, 5 at 1b, 6 at 4b, 9 at 12b, 12 at 27b. Where the single final layer disagrees, it is called out explicitly — see the read-out caveat. Coverage: 14 / 15 / 14 / 14 / 14 lens points (270m / 1b / 4b / 12b / 27b) — all five complete. **Behavioral: all five arms are judge-scored, complete, and on one instrument** (14 / 15 / 14 / 14 / 14 points incl. $t=0$), so cross-size behavioral comparison is legitimate for the first time in this deck.

The probe step in detail

Runs in `.venv-jlens` (transformers 5.14.1 + jlens @ 581d398) — a second env, because the training pin 4.53.2 is hard-incompatible. Imports nothing from `gemma_med`.

Per checkpoint (base prepended as step 0, then `checkpoint-*`):

1. **Load** → `jlens.from_hf(hf, tok)`. Forced text-only Gemma3ForCausalLM; any loader leaving a weight missing is *rejected*, not accepted.
2. **Fit** a fresh lens: `jlens.fit(model, corpus, dim_batch=16, max_seq_len=128)` on the frozen 1000-seq WikiText corpus. *Dominates runtime*.
3. **Concordance**: for each of the 193 probes, `lens.apply(model, prompt, positions=[-1])` → per source layer, top-1 agreement and $\text{KL}(\text{model} \parallel \text{lens})$ at the last token.
4. **Drift** vs. $t=0$, per layer: relative Frobenius $\|J_l - J_l^0\| / \|J_l^0\|$ and cosine.

Output. One `fsync`'d JSON row per checkpoint in `metrics.jsonl`, plus W&B scalars keyed by training step. Resumable: completed steps are skipped and the fit checkpoints every 50 prompts, so a requeue does not wipe the trajectory.

Invariants that make the trajectory comparable

- **One load path for all points.** Letting `Auto*` pick per-path would read $t=0$ through the multimodal wrapper and later points through the text model — two forwards on one trajectory, silently corrupting drift.
- **Lens refit every time.** J is a function of the weights.
- **Corpus and probe set frozen** across all checkpoints and sizes.
- **$t=0$ lens cached in fp32**, so a resumed run's drift baseline matches a fresh run's exactly.

Reading the metrics — the two drift measures

J_l = lens Jacobian at source layer l ; J_l^0 = the same layer in the $t=0$ base lens. n = number of source layers (18 / 26 / 34 / 48 / 62 at 270m / 1b / 4b / 12b / 27b). Both measures are computed per layer, then averaged over all n ; both are exactly 0 / 1 at $t=0$ by construction.

Metric	How it is computed	Direction
rel. Frobenius relfro_mean	$\ J_l - J_l^0\ _F / \ J_l^0\ _F$ — the <i>size</i> of the change, normalised by the base magnitude.	Neither better nor worse. It says <i>how far</i> the lens moved, not whether the move helped. Larger = more change.
cosine cos_mean	$\langle J_l, J_l^0 \rangle / (\ J_l\ \ J_l^0\)$, flattened over all entries — the <i>direction</i> of the change.	Neither. 1 = pointing the same way as the base; lower = more rotation. Bounded above by 1.

Why both. They can move independently: a lens can grow in magnitude without rotating (relfro up, cosine ≈ 1) or rotate without changing size. In practice they track each other here, which is itself a mild sanity check on the fit.

Not comparable across sizes. J_l has a different shape at every arm, so only these normalised scalars travel — never a raw Jacobian entry.

Reading the metrics — the four faithfulness measures

All four compare the **lens read-out** against the **model's actual next-token distribution**, at the last prompt token, averaged over the prompts of one cue. They differ in which layers they read and how they score.

Metric	How it is computed	Direction
faithfulness KL kl_final	KL(model lens), averaged over the last $n/5$ layers . <i>The workhorse metric.</i>	Lower is better — the lens predicts what the model actually says. 0 = perfect; unbounded above.
top-1 agreement top1_final	Fraction of prompts where $\arg \max \text{ lens} = \arg \max \text{ model}$, same last- $n/5$ layers.	Higher is better , range 0–1. Much coarser than KL — see caveats.
top1_auc	The same agreement averaged over all n layers , not just the top fifth. Despite the name, a mean rather than an integral.	Higher is better — agreement holds deeper down the stack, not only near the output.
crossover_frac	Shallowest layer whose top-1 agreement reaches 0.5, as a fraction of depth; 1.0 if it never does.	Lower is better — the lens locks onto the model's answer earlier in depth.

Cues: `general` / `medical` are the 50+50 hand-authored style-matched pairs; `multihop` the 93 externally sourced prompts. *The cue is the unit of comparison* — `medical` is read against `general`, never on its own. This deck reports `kl_final` and `top1_final`; the other two are computed and stored but not used.

Reading the metrics — direction, and what not to do

Behavioral metrics

- **answer rate** — % of generations that committed to an option. **Higher better.** A pure *format* measure.
- **answered accuracy** — accuracy among those only. **Higher better.** **This is the knowledge signal.**
- **raw accuracy** — non-answers counted wrong. **Higher better**, but confounded: it moves when *either* knowledge or formatting moves.

Why it matters: on 270m PubMedQA, raw accuracy rises 12.6 → 43.6 — reading as a large gain. Answered accuracy over the same span *falls*, 57.8 → 55.0: all of it is the answer rate going 21.8% → 79.2%. The model learned to *commit to an option*, not to answer correctly. **Raw alone inverts the sign.**

Directional summary. Drift (relfro ↑, cosine ↓) says the lens *changed*; KL ↑ and top-1 ↓ say it *got worse at predicting the model*. The two are independent — a lens can move far and stay faithful, or barely move and lose faithfulness.

Three rules for reading these numbers

1. **Never compare absolute KL across sizes.** `kl_final` averages a *different number of layers* per arm (3 / 5 / 6 / 9 / 12), over models of different depth and width. Only the **within-arm Δ from $t=0$** is comparable across arms.
2. **Always read against $t=0$.** Every figure carries the untuned base as a dashed line or a pale bar. A finetuned-only number says nothing.
3. **Name the read-out.** “KL” is ambiguous: the layer mean and the single final layer disagree in sign on 270m medical. This deck is the **layer mean** throughout.

Findings

1. **Drift has a sharp onset, then saturates** — and at 4b/12b partly *retreats*. The Jacobian finishes moving in the first ~ 100 steps. **Onset gets earlier with size**; at 12b it is underway by step 2.
2. **The faithfulness cost is not monotone in size — it peaks at 4b and vanishes.** ΔKL on the general control runs +0.09/ +0.54/ +2.08/ +1.29/−0.25: an inverted U. 12b was the descending limb, not an anomaly, and **27b pays no cost at all**.
3. **Medical is protected, increasingly so with size.** ΔKL medical −0.05/ −0.37/+1.19/ −1.98/−6.26, with top-1 agreeing. **4b is the sole outlier on both cues** across five points, and why is unexplained. **Both findings from the three-arm deck are now retracted outright: the cost does not scale with size, and medical is not unprotected.**
4. **Medical SFT works, but the gain is bounded by headroom, not size.** Answered-accuracy Δ on MedQA / MedMCQA / PubMedQA: 270m −1.9/ −10.2/ −2.8, 1b +4.3/ +2.7/ +2.8, 4b +8.2/ +7.0/ +13.2, 12b +1.0/ +6.2/ +21.6. **270m is the only arm that gets worse.**
5. **The timing ordering holds; the correlation behind it does not.** Drift finishes (steps 16–32) long before accuracy moves (2048+) at every size — but drift-vs-answer-rate is only −0.33 to −0.48 on MedQA, and +0.92 at 270m on MedMCQA. **The $r \approx -0.9$ in the previous deck was an artifact of $r_p=1.0$.**

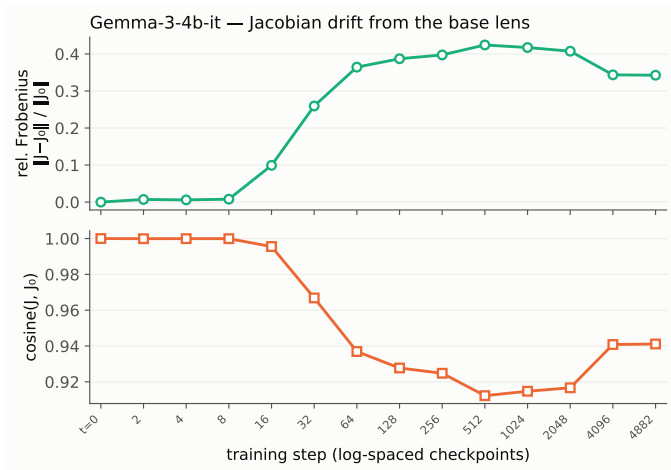
1. Drift onsets sharply, then saturates

	onset (mean > 0.05)	peak relfro	final relfro / cos (layer mean)	per-layer max
270m-it	step 64	0.250 @ 1024	0.249 / 0.976	0.350
1b-it	step 16	0.308 @ 2048	0.279 / 0.956	0.418
4b-it	step 16	0.424 @ 512	0.343 / 0.941	0.531
12b-it	step 2	0.472 @ 2048	0.421 / 0.897	0.692
27b-it	step 2	0.466 @ 2048 [†]	0.466 / 0.874	0.697

- **Onset is earlier and the swing larger the bigger the model**, now over five sizes: final relfro $0.249 < 0.279 < 0.343 < 0.421 < 0.466$, and cosine falls monotonically $0.976 > 0.956 > 0.941 > 0.897 > 0.874$. The one size trend every added arm *strengthens* rather than breaks. Cosines are **fp64-corrected** (see caveats) — the old read-out's bias grew with d_{model} , so correcting it steepens this trend.
- The “nothing moves for ~8 steps” claim is now false, and 27b says why. It held at ≤ 0.008 for 270m/1b/4b; 12b reads 0.055 at step 2 and 27b 0.060 — two independent d_{model} reproducing it. That favours **real early motion at large models** over a noisy fit at 3840 specifically, which was the open alternative.
- The late partial retreat does *not* generalise upward. 4b walks back 19% from its 512 peak, 12b 11% from 2048, 1b 9%, 270m flat — but **27b plateaus without retreating** (0.466/0.465/0.466 at 2048 / 4096 / 4882). The retreat is a middle-sizes phenomenon, not a large-model one.

Robust under either drift measure. [†]27b's last three points are tied to ± 0.0006 — a plateau, not a late climb.

1. Drift — 4b-it



The cosine panel is **fp64-corrected**. It previously read *above* 1.0 at steps 2–8 — not a small-drift curiosity but the visible tip of a bias present at every step, worth $\sim 0.26\text{pp}$ here and growing with d_{model} . See caveats.

2–3. Faithfulness vs. size — **not monotone; it turns over at 4b**

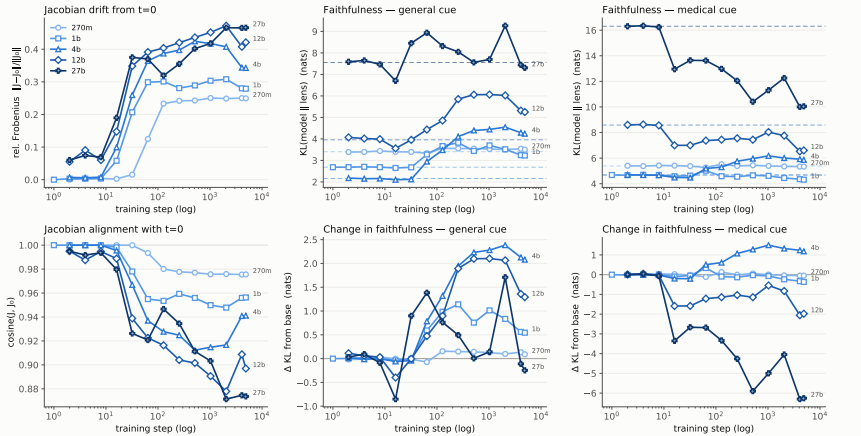
Lens↔model KL, mean over the last $n/5$ layers (the probe's `kl_final`) — *lower = lens predicts the model better*. Base → final checkpoint.

	ΔKL general			ΔKL medical		ΔKL multihop (OOD)									
270m-it	3.407	→ 3.497	(+0.09)	5.388	→ 5.340	(−0.05)	1.958	→ 1.975	(+0.02)						
1b-it	2.684	→ 3.229	(+0.54)	4.682	→ 4.314	(−0.37)	2.423	→ 3.430	(+1.01)						
4b-it	2.163	→ 4.243	(+2.08)	4.683	→ 5.867	(+1.19)	4.088	→ 5.715	(+1.63)						
12b-it	3.959	→ 5.251	(+1.29)	8.584	→ 6.606	(−1.98)	7.267	→ 9.578	(+2.31)						
27b-it	7.555	→ 7.308	(−0.25)	16.308	→ 10.049	(−6.26)	11.782	→ 15.344	(+3.56)						
top-1 medical:	270m	0.260	→ 0.260	1b	0.448	→ 0.572	4b	0.677	→ 0.660	12b	0.596	→ 0.720	27b	0.587	→ 0.737

- **“Monotone in size” is dead on general**: +0.09/ +0.54/ +2.08/ +1.29/−0.25 is an **inverted U peaking at 4b and crossing zero by 27b**. 12b was the descending limb, not an anomaly. multihop alone stays monotone — **but on the layer mean only; at 27b its final layer reads −0.02**.
- **“Medical is not protected” is dead too**, oppositely: −0.05/ −0.37/+1.19/ −1.98/−6.26 — protection **deepens with size**, top-1 agrees (+0.150 at 27b, the largest of any arm), and **4b is the sole outlier on both cues**.
- **The real result: drift and cost decouple**. relfro rises monotonically with size (0.249 → 0.466) while the general-cue cost falls to nothing — **27b moves most and pays least**. Movement is not infidelity; the glossary asserted that, this measures it.
- **Not settled**: −6.26 nats is partly **headroom** — 27b starts least faithful of all (7.55 / 16.31), and against its own base the protection series is −0.9/ −7.9/+25/ −23/−38%. An inverted U is also what **one fixed LR** would draw.

2–3. All five arms on one axis

Gemma-3 -it · medical SFT · j-lens trajectory by model size · KL = mean over the last n/5 source layers · dashed = that arm's base (t=0)



Arms shaded light→dark by size (an *ordinal* ramp — size is ordered, so it does not take categorical hues). x is the real training step on a log axis, since the arms have different step grids; $t=0$ is each arm's dashed baseline (top row) and the zero line (bottom row), never a point. **The bottom-right panel is the held finding:** 12b (darkest) runs *below* zero for the whole trajectory while 4b runs above it.

2–3. What survives all five arms

Both headline size claims reversed, but the trajectory data are not uninformative. One statement holds at **every** size:

	Δ medical	Δ general	Δ multihop	medical is...
270m-it	−0.05	+0.09	+0.02	least damaged
1b-it	−0.37	+0.54	+1.01	least damaged
4b-it	+1.19	+2.08	+1.63	least damaged
12b-it	−1.98	+1.29	+2.31	least damaged
27b-it	−6.26	−0.25	+3.56	least damaged

- **Medical is the least-damaged cue at all five sizes** — by 0.14/0.91/0.90/3.27/6.01 nats over general, a margin that **widens with size**. The only claim that has held from two arms to five, and it is *directional* — so it does not rest on the cross-size absolute comparison the deck forbids.
- **Medical SFT buys the medical cue relative ground at every size**. Whether it also buys *absolute* ground is what flips: yes at 270m/1b/12b/27b, no at 4b — so on five points the absolute claim holds everywhere *except* 4b.
- **Not “medical improves” generally** — at 4b it degrades; the universal claim is only that it degrades *less*. 27b does both.

Read-out check. Both cues this slide rests on agree in sign across read-outs at every size but 270m. 27b multihop flips (+3.56 mean vs. −0.02 final) — two flips in five arms; see the caveat.

2–3. The domain gap narrows at every size — by three mechanisms

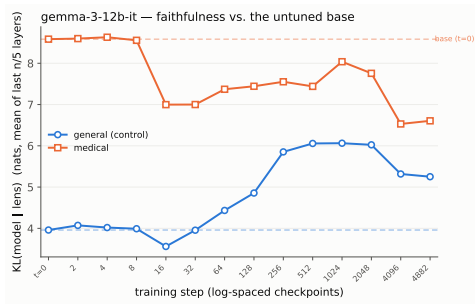
Within each arm the medical cue starts 2–8.8 nats less faithful than general. **All five arms narrow that gap** — but not by the same mechanism.

	$t=0$	final	Δ	closes by
270m-it	1.981	1.843	-0.138	both ends
1b-it	1.998	1.085	-0.912	both ends
4b-it	2.520	1.624	-0.895	general only
12b-it	4.625	1.355	-3.271	both ends
27b-it	8.753	2.741	-6.012	both ends

At 270m, 1b and 12b the gap closes **from both ends** — medical improves *and* general regresses toward it.

At 4b it closes from one end: both cues degrade, general just faster (+2.08 vs. +1.19). **A -0.9 gap change means opposite things at 1b and 4b** — which is why the gap is never reported on its own.

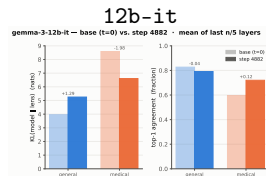
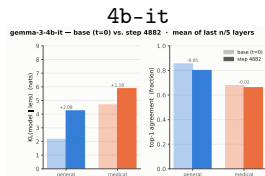
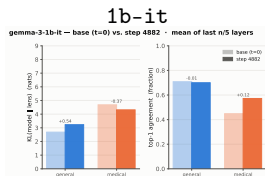
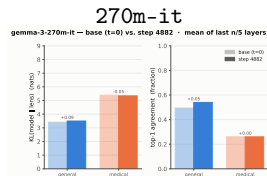
27b closes the widest gap (8.75) by 69%, from both ends. multihop degrades at *all five* sizes.



The 12b panel: medical (upper) falls hard toward general (lower), which rises — the “both ends” shape.

Contrast the 4b panel, where both curves rise away from their dashed $t=0$ baselines. All five arms are on the previous slide.

2–3. Base vs. finetuned, side by side



Pale bar = untuned base ($t=0$), solid = step 4882. All four are pinned to the same axes (KL 0–9 nats, top-1 0–1), so the panels are directly superimposable — the range widened from 0–6 to fit 12b's medical base at 8.58. Compare the Δ labels, not the bar heights: absolute KL is not comparable across sizes (different depth and width, and a different number of layers in the $n/5$ mean). Read that way, 270m's largest KL move (+0.09) is about a quarter of 1b's smallest (−0.37), and 12b's medical move (−1.98) is the largest at any size in either direction.

4. The instrument, and why every old behavioral number is void

Two things changed at once on 2026-08-05, and both invalidate comparisons to any earlier version of this deck.

1. Decoding: `rp=1.0` → 1.1

- The old answer-rate collapse was **verbatim repetition loops under greedy decoding**, not chain-of-thought overrunning the token cap. At 4b step-4882, 311 of 327 unanswered MedMCQA rows were loops; 8 were real truncation.
- A repetition penalty fixes it outright: 4b MedQA answer rate now never leaves 97.9–99.8% across the whole trajectory (it bottomed at 83.8% before).
- **No capability appears.** `acc_answered` is essentially unchanged — the knowledge was never missing, only the format.

Both changes are cross-validated. Judge: on 300 identical MedQA rows at 4b step-1024, Claude Haiku and Kimi agree to within 1.3 pp raw (45.0 vs. 46.3). Judge decode: re-judging the same rows twice flipped chosen on 1.34% under the old wire vs. 0.33% at $t=0$, so the old config moved `acc_raw` by up to 1.0 pp *between two identical runs*; old-vs-new differs by ≤ 0.67 pp, i.e. inside that noise. Freezing buys reproducibility, not a different answer. $t=0$ is still not bitwise deterministic

2. Judge: Claude Haiku → self-hosted Kimi

- Scoring is an **LLM judge** (`gemma_med.judge`), not a regex: it matches option *content*, not the letter, and is told the gold answer and forbidden to re-derive it.
- Judge *decode* is now frozen too — `temperature=0.0`, `seed=0`, recorded as `judge_decode`. Before this date each provider sampled at its own default, so “the judge” was frozen at the model level only.
- **All five arms are now judge-scored by the same judge**, which is what makes the cross-size table on the next slide legitimate.

4. Behavior — all five arms, one instrument

Judge-scored at $rp=1.1$. **Answered** accuracy, with answer rate in parentheses. $t=0$ is each arm's untuned base.

	MedQA		MedMCQA		PubMedQA	
270m-it $t=0$	26.7	(96.5%)	36.9	(59.3%)	57.8	(21.8%)
270m-it final	24.8	(89.5%)	26.6	(92.5%)	55.0	(79.2%)
Δ	-1.9		-10.2		-2.8	
1b-it $t=0$	31.1	(99.9%)	34.3	(99.9%)	51.6	(100%)
1b-it final	35.3	(99.0%)	37.0	(99.0%)	54.4	(99.6%)
Δ	+4.3		+2.7		+2.8	
4b-it $t=0$	50.9	(99.9%)	46.6	(99.8%)	60.0	(100%)
4b-it final	59.2	(99.5%)	53.6	(99.5%)	73.2	(100%)
Δ	+8.2		+7.0		+13.2	
12b-it $t=0$	70.0	(100%)	58.8	(100%)	56.4	(100%)
12b-it final	71.0	(98.8%)	65.0	(99.5%)	78.0	(100%)
Δ	+1.0		+6.2		+21.6	
27b-it $t=0$	77.1	(100%)	63.6	(99.9%)	61.6	(100%)
27b-it final	77.3	(99.6%)	64.4	(98.6%)	79.2	(100%)
Δ	+0.1		+0.9		+17.6	

- **270m is the only arm that gets worse** — not by format churn: its answer rate *improves* on two of three while answered accuracy falls on all three.
- **The gain is bounded by headroom, not size.** 27b: +0.1/ +0.9 from bases of 77.1/63.6, but +17.6 on PubMedQA from 61.6 — **the largest arm gains least where it knows most.**

4. A second failure mode: collapse onto the majority label

Format collapse is not the only thing that goes wrong mid-training. Share of *answered* items assigned to the dominant label, against that label's prevalence in the gold answers:

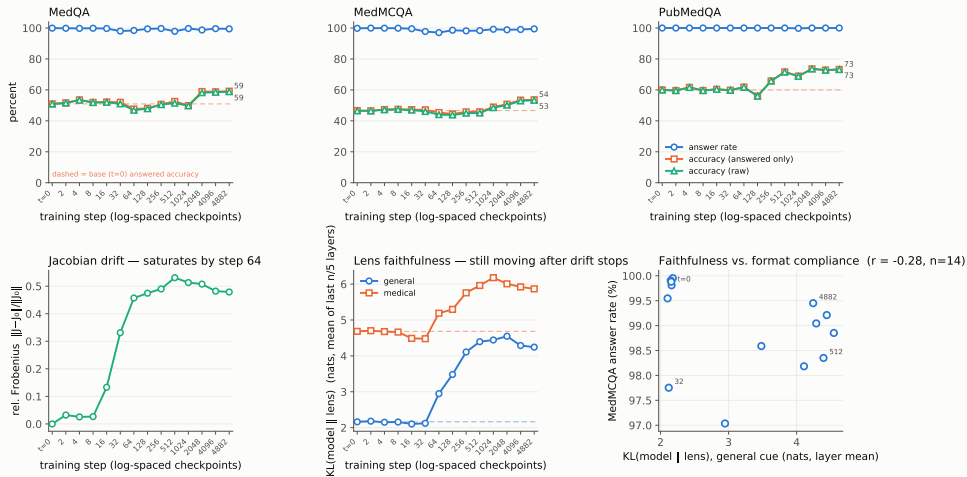
	MedQA ("A")		MedMCQA ("A")		PubMedQA ("yes")	
	chosen	gold	chosen	gold	chosen	gold
$t=0$	18.5	27.7	21.6	32.2	50.4	55.2
step 512	28.6	27.7	37.5	32.2	65.2	55.2
step 1024	37.7	27.7	39.3	32.2	69.7	55.2
final	29.5	27.7	33.1	32.2	66.8	55.2

- The base model is *anti*-majority (18.5 vs. 27.7). SFT pushes it through neutral and overshoots.
- Step 1024 is a spike, not a trend — it recovers. Worth knowing before building a story on any single checkpoint.
- PubMedQA is the control that reframes it. It has no "A" to collapse onto, yet shows the same overshoot on "yes". So this is **majority-label over-commitment, not position bias**.

This mode survives the instrument change — recomputed at $rp=1.1$, it is the same shape, roughly two-thirds the amplitude (the MedQA step-1024 spike was 57.2 at $rp=1.0$, 37.7 now). Unlike the format-collapse mode, it is not a decoding artifact. Note this means the two failure modes no longer trade off: with format compliance held near 100% throughout, over-commitment is now the *only* mid-training pathology visible at 4b.

4. Joined behavioral + lens trajectory — 4b-it

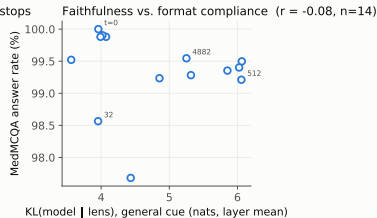
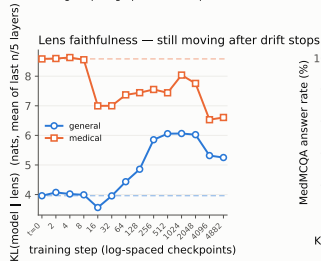
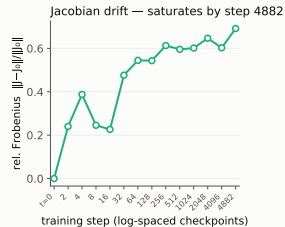
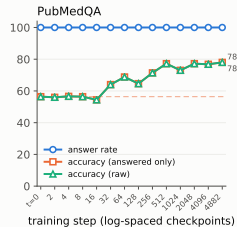
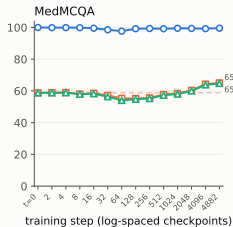
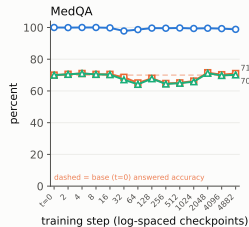
gemma-3-4b-it · medical SFT · behavioral trajectory vs. Jacobian lens



Judge-scored at $rp=1.1$. Compare this against the previous version of the deck: answer rate (blue) now sits flat near the top instead of diverging from answered accuracy (orange) at step 32. The format/knowledge split that this panel used to make visible was the decoding.

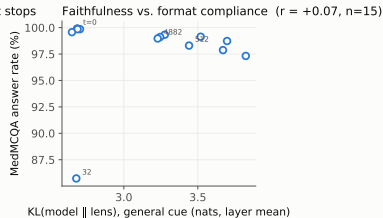
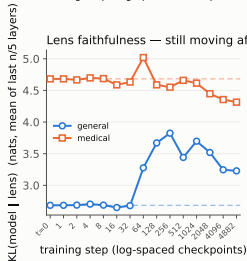
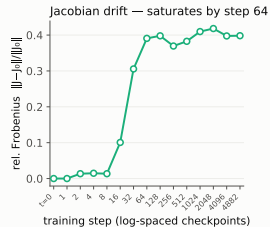
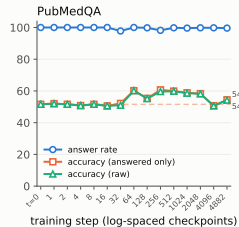
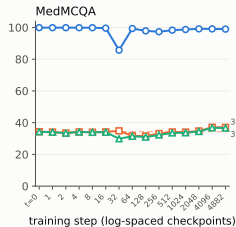
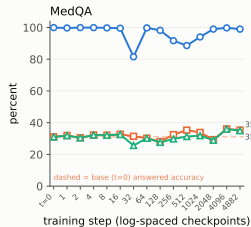
4. Joined behavioral + lens trajectory — 12b-it

gemma-3-12b-it · medical SFT · behavioral trajectory vs. Jacobian lens



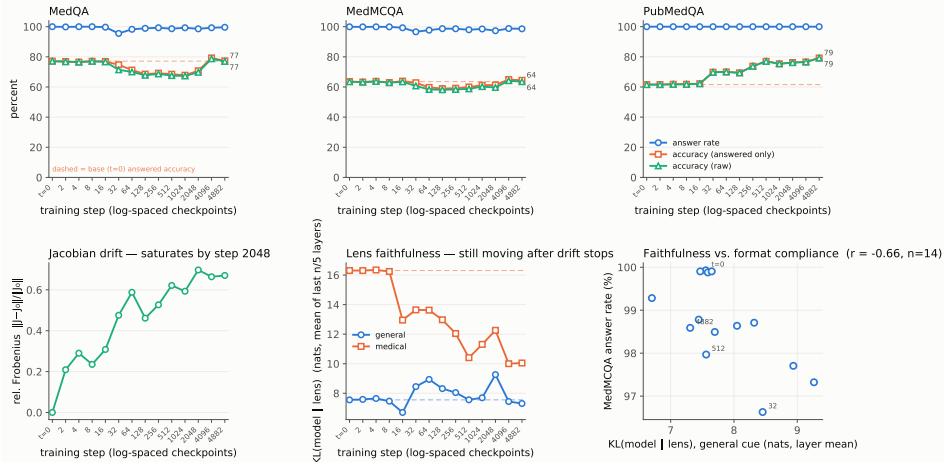
4. Joined behavioral + lens trajectory — 1b-it

gemma-3-1b-it · medical SFT · behavioral trajectory vs. Jacobian lens



4. Joined behavioral + lens trajectory — 27b-it

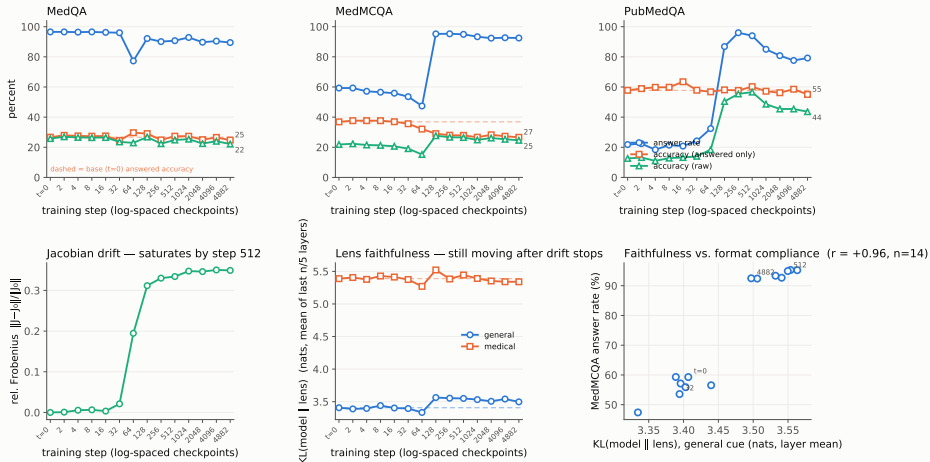
Gemma-3-27b-it · medical SFT · behavioral trajectory vs. Jacobian lens



The one arm that dips and recovers. MedQA and MedMCQA fall ~9pp to a trough at steps 128–1024, then return to base by 4096; PubMedQA rises throughout. **So its Δ of +0.1/+0.9 is a round trip, not a flat line** — read the table slide with that in mind.

4. Joined behavioral + lens trajectory — 270m-it

gemma-3-270m-it · medical SFT · behavioral trajectory vs. Jacobian lens



The KL panel is the layer mean, as everywhere else in this deck; at the single final layer the medical curve rises instead. 270m is the one arm where answer rate still moves a lot under $rp=1.1$ — it *rises* (MedMCQA 59 $\rightarrow$ 92%), which is why its drift $\leftrightarrow$ rate correlation comes out *positive*.

5. Timing holds; the correlation behind it does not

The ordering is the robust part, and it holds at all five sizes — 4b-it shown:

Signal	Onset	Notes
Jacobian drift	steps 16–32	0.099 → 0.259 (layer mean); plateaus by step 64
Task accuracy	steps 2048+	MedQA answered 52.6 at step 512; 59 only at 2048–4882

But “drift leads format collapse” does not survive $rp=1.1$ — Pearson r vs. drift (layer mean), answer rate / answered accuracy:

	MedQA	MedMCQA	PubMedQA
270m-it	−0.49 / −0.22	+0.92 / −0.99	+0.97 / −0.56
1b-it	−0.33 / +0.11	−0.20 / +0.05	−0.31 / +0.73
4b-it	−0.47 / +0.06	−0.65 / +0.15	−0.27 / +0.57
12b-it	−0.48 / −0.49	−0.48 / +0.07	— / +0.90
27b-it	−0.38 / −0.45	−0.73 / −0.31	— / +0.92

- The old −0.91 on 4b MedMCQA is now −0.65, and the sign **flips outright at 270m** (+0.92); **27b adds no support** (−0.38 on MedQA). (PubMedQA at 12b/27b is --: answer rate is exactly 100.0% at every checkpoint, so r is undefined.)
- **With format compliance pinned near 100% by the decoding, there is almost no variance in answer rate left to correlate with.** The old r was measuring an artifact of greedy decoding.
- **What still stands:** the Jacobian finishes moving before the model learns any medicine. **What does not:** that it moves *with* format collapse specifically.

Caveats — the measurements

- **Top-1 is coarse.** $n=50$ prompts for general/medical (93 for multihop), so *per-layer* top-1 quantizes at 0.02; the reported statistic averages 3 / 5 / 6 / 9 / 12 layers, giving a 1/150–1/600 grain that makes small moves look more precise than they are. Single-checkpoint wiggles of ± 0.04 are still noise. *Trust the KL trends, not top-1.*
- **Read-out choice is not innocent.** Layer-mean and final-layer KL disagree in sign on 270m medical. Every claim here uses the layer mean; an earlier draft of this deck used the final layer and reported a 270m $\leftrightarrow$ 1b *inversion* that does not survive. Report the statistic, not just the number. **It recurs at 27b** — multihop reads +3.56 on the layer mean and -0.02 at the final layer, so the still-monotone-multihop reading is layer-mean-only. Two flips in five arms; neither is on a cue carrying a headline claim, but the rate is not zero.
- **The drift cosine was biased — diagnosed and fixed (2026-08-10).** `jac_drift` divided an fp32 `torch.dot` (accurate to 6×10^{-6}) by fp32 `torch.linalg.vector_norm`, which *underestimates* $\|J\|$ by 0.69% at 27b: it accumulates d_{model}^2 terms until the smallest are absorbed. **An earlier draft of this slide blamed dot and called it harmless — both wrong.** Diagnostic: $\cos(J_0, J_0)$, exactly 1 by definition, evaluated to 1.0138. The bias is positive at every step and size and grows with d_{model} ($0.01 \rightarrow 0.02 \rightarrow 0.26 \rightarrow 0.47 \rightarrow 1.4\text{pp}$ for 270m $\rightarrow$ 27b), so it was a **cross-size** bias inside a cross-size claim (finding 1). `jac_drift` now accumulates in fp64; 270m–12b are retro-corrected in place (`scripts/correct_drift_cos.py`, raw kept as `cos*_fp32raw`), 27b is not yet. Finding 1 survives and steepens; nothing else in the deck reads the cosine. `relfro` is left uncorrected — both its terms are norms, so the bias largely cancels and $\sim +0.1\%$ relative survives at 4b.

Caveats — the scope

- **Five sizes is still not a scaling curve.** 27b settled *which* shape the general cue has — an inverted U peaking at 4b — but five points spanning $\sim 100\times$ in parameters is four intervals, and the turning point rests on **one arm** (4b), which is also the one unexplained outlier. **A shape identified from its own outlier is a hypothesis, not a scaling law.**
- **One hyperparameter setting — now the load-bearing caveat.** All five arms are full FT at $\text{lr } 1\text{e}-5$ on one mixture. Whether these deltas are about *size*, or about a fixed LR acting differently as the model grows, is not separable from these runs — and **an inverted-U general curve is exactly that confound's signature.** Adding 27b did not weaken this caveat; it sharpened it. The decisive experiment is a second LR at one size, not a sixth size.
- **The behavioral instrument is finally uniform** — five arms, one decode config, one judge, judged $t=0$ baselines for all. Cross-size behavioral comparison is legitimate for the first time. **The cost:** every behavioral number predating 2026-08-05 is void, and finding 5 did not survive the change.
- **Coverage gaps.** 4b, 270m and 12b have no checkpoint-1, so their grids start at step 2; only 1b probed step 1. No -pt arm exists at any size — the pt-vs-it starting-point axis is entirely unstarted.
- **The 12b early-step drift is unexplained.** 0.055 at step 2 against ≤ 0.008 for the three smaller arms. Real early motion or a noisier fit at $d_{\text{model}}=3840$ — not yet distinguished, and it is the one number in finding 1 that is provisional.

Next

1. **A second LR at one size** — 4b or 12b at, say, $3e-6$. **The highest-value experiment in the study**, and more so now: the inverted U is settled as a *shape*, but an inverted U is also exactly what “a fixed $lr\ 1e-5$ acting differently at different scales” would draw. Finding the turnover does not disambiguate it — only a second LR does.
2. **Explain 4b** — across five points it is the only arm that loses faithfulness on the medical cue, and the peak of the general cost. Re-verify its trajectory before the anomaly is theorised about: it is the smallest size on the Gemma3ForConditionalGeneration load path, so a provenance check comes before an explanation.
3. **Widen the general/medical cues** beyond the 100 hand-authored prompts, to check the domain-gap collapse is not an artifact of how those pairs were written. 12b’s 4.63-nat starting gap makes this more pressing, not less.
4. **Open a -pt arm** at 1b: does a non-aligned starting point show the same drift-before-competence ordering?
5. **Anchor against medgemma-27b-text-it** on the same probe set to place the trajectory endpoints on an absolute scale.

Answered since the last version: **27b exists end to end**, so findings 2–3 are settled rather than held. **Nothing newly retracted** — the three standing retractions (size-monotone cost, medical-is-not-protected, drift-leads-format-collapse at $r = -0.91$) were made by 12b and the decode fix; 27b only confirms the first two.

Data provenance

Artifact	Path
Lens metrics	eval/jlens/{270m_it_full, 1b_it_full_v3, 4b_it_full, 12b_it_full}/metrics.jsonl (14/15/14/14 rows)
Joined traj. $t=0$ dirs	eval/traj/<arm>_rp11/joined.json (all judge-scored) eval/gemma-3-{270m,1b,4b,12b}-it-baseline-rp11/
Fit corpus	data/jlens/fit_corpus_v2.txt (1000 WikiText seq, frozen)
Probe set	data/jlens/probe_set_v2.tsv (50/50/93, frozen)
Figures	figs/<lens-tag>/ lens (decode-independent); figs/<arm>_rp11/fig_joined.pdf (decode-dependent)
Probe jobs	25776556 (1b), 25799677 (270m), 25948088-25948100 (4b), 13-way fanout (12b)

All four runs logged fit_corpus=1000 probes=193 — the frozen v2 inputs. The v1 files (40 seq / 30 prompts) are legacy, still in data/jlens/, used only by the superseded runs.

4b and 12b were probed one Slurm job per checkpoint (4b ~3.5 h each; 12b longer — fit cost scales as $d_{\text{model}} \times \text{params}$), all 13 workers reusing the *same* fp32 $t=0$ lens so drift shares one baseline; rows merged by scripts/merge_jlens_fanout.py, which refuses the merge if any non-base step appears twice.

Superseded: 4b_it_validate{,2} (wrapper-validation spikes), 1b_it_full_v2 (v1 30-prompt set), 1b_it_full_probev2, 270m_verify (Phase-0 smoke test).

Data provenance — the behavioral instrument (all five arms)

Field	Value
Decode	temperature=0.0, max_new_tokens=1024, repetition_penalty=1.1, seed=0 frozen 2026-08-05
Scorer	<code>gemma_med.judge</code> (replaces <code>answer_parsing.py</code>)
Provider	local — the lab's self-hosted Kimi, BigPurple only
Judge model	Barney frozen for all five trajectories
Judge decode	temperature=0.0, seed=0 also frozen 2026-08-05
Verdicts	correct / incorrect / no_answer, schema-constrained
Judgments	<code>eval/traj/*_rp11/step-*/*_judgments.jsonl</code>
Provenance	<code>.../step-*/judged_summary.json</code> (13/14/13/13/13 files)

A second decode is on disk and must not be mixed in. Every arm was also decoded at `rp=1.05` (`eval/traj/*_rp105/`), kept as a sensitivity check: it halves the repetition loops while costing less `acc_answered` than 1.10 does, and it preserves a muted version of the old answer-rate dip (4b MedQA 92.9% at step 512 vs. 97.9% at `rp=1.1`). **Different instrument — never in the same figure.**

`judge_self_disagreement` — **a quality signal that scores nothing.** It counts rows where the judge's verdict string conflicts with the option it picked: 0.05–0.25% on these runs. **It changes no number** — correct and answered are both recomputed from chosen, never from verdict. It ran higher (0.95% overall, peaking 2.8%) on the `rp=1.0` corpus, where the judged model rambled more; the statistic tracks **how ramblly the judged model is**, not how confused the judge is.